



SOP #17

5XTBS & 1XTBS-T

Tris-buffered saline (abbreviated TBS) is a buffer used in some biochemical techniques to maintain the pH within a relatively narrow range.

TBS has many uses because it is isotonic and non-toxic.

TBS can be made in a stable, 5X stock solution that resists microbial growth at room temperature because of the high salt content.

This 5X stock must be pH'd to 7.6 before using, otherwise the Western Blot will not work. When you pH, add a note to the label to confirm (along with your initials and the date).

Tween-20 is added in order to use it as a blocking & washing solution in Western Blotting. The Tween acts as a surfactant, and it is very viscous. We kept a 10% solution in the lab to make adding volumes to the stock 5X TBS easier. The final concentration of Tween is 0.01%.

IMPORTANT NOTE: The 1XTBS-T is prone to growing microbial contaminants. These fungal and bacterial contaminants can eat the blocking powder as well as the antibody and protein targets on the membrane! Never use cloudy buffer or buffer with floating things in it - always re-make fresh buffer. Bottles should be washed carefully with soap between batches, and preferably dried upside down.

5X TBS

40 g NaCl
12.1 g Trizma® Base
Make to 1.0 L with ddiH₂O
pH to 7.6

1XTBS-T (0.1%)

10 ml 10% Tween-20
200 ml 5X TBS
790 ml ddiH₂O